In [ ]:
```
EXPERIMENT 2 – NUMPY OPERATIONS
PIN NO:- 241UDS0111
AIM:-
To study and practice NumPy operations including array creation, attributes, indexing, slicing, broadcasting,
arithmetic, statistical operations, concatenation, reshaping, sorting, splitting, and retrieving selected data.
```

In [ ]:
```
## Step 1 – Install and Import NumPy
If NumPy is not installed, run `%pip install numpy pandas` in a separate code cell.
```

In [1]:
```
import numpy as np
import pandas as pd
```

In [ ]:
```
## Step 2 – Create Arrays for Three Facilities
Each array contains Week 1–Week 4 energy consumption in kWh.
```

In [2]:
```
library_energy_241UDS0111 = np.array([860, 915, 890, 940])
computer_lab_energy_241UDS0111 = np.array([1240, 1315, 1275, 1360])
sports_centre_energy_241UDS0111 = np.array([720, 780, 755, 810])

library_energy_241UDS0111, computer_lab_energy_241UDS0111, sports_centre_energy_241UDS0111
```

Out[2]:
```
(array([860, 915, 890, 940]),
 array([1240, 1315, 1275, 1360]),
 array([720, 780, 755, 810]))
```

In [ ]:
```
## Step 3 – Check Array Properties
```

In [3]:
```
library_energy_241UDS0111.ndim
```

Out[3]:
```
1
```

In [4]:
```
computer_lab_energy_241UDS0111.shape
```

Out[4]:
```
(4,)
```

In [5]:
```
sports_centre_energy_241UDS0111.size
```

Out[5]:
```
4
```

In [6]:
```
library_energy_241UDS0111.dtype
```

Out[6]:
```
dtype('int64')
```

In [7]:
```
library_energy_241UDS0111.itemsize, computer_lab_energy_241UDS0111.itemsize,
sports_centre_energy_241UDS0111.itemsize
```

Out[7]:
```
(8, 8, 8)
```

In [ ]:
```
## Step 4 – Retrieve Data Using Indexing and Slicing
```

In [8]:
```
sports_centre_energy_241UDS0111[0]
```

Out[8]:
```
np.int64(720)
```

In [9]:
```
np.unique(computer_lab_energy_241UDS0111)
```

Out[9]:
```
array([1240, 1275, 1315, 1360])
```

In [10]:
```
library_energy_241UDS0111[0:2]
```

Out[10]:
```
array([860, 915])
```

In [11]:
```
computer_lab_energy_241UDS0111[2:]
```

Out[11]:
```
array([1275, 1360])
```

In [12]:
```
sports_centre_energy_241UDS0111[-3:]
```

Out[12]:
```
array([780, 755, 810])
```

In [ ]:
```
## Step 5 — Assign New Values Using Broadcasting
Copy the library array, append two more weekly values, then assign updated values to the last two positions.
```

In [13]:
```
library_energy_new_241UDS0111 = library_energy_241UDS0111.copy()
library_energy_new_241UDS0111
```

Out[13]:
```
array([860, 915, 890, 940])
```

In [14]:
```
library_energy_new_241UDS0111 = np.append(library_energy_new_241UDS0111, [980, 955])
library_energy_new_241UDS0111
```

Out[14]:
```
array([860, 915, 890, 940, 980, 955])
```

In [15]:
```
library_energy_new_241UDS0111[4:6] = [1000, 970]
library_energy_new_241UDS0111
```

Out[15]:
```
array([ 860,  915,  890,  940, 1000,  970])
```

In [ ]:
```
## Step 6 — Arithmetic Operations
```

In [16]:
```
sports_centre_energy_241UDS0111
```

Out[16]:
```
array([720, 780, 755, 810])
```

In [17]:
```
sports_centre_energy_241UDS0111 + 25
```

Out[17]:
```
array([745, 805, 780, 835])
```

In [18]:
```
sports_centre_energy_241UDS0111[3] - 15
```

Out[18]:
```
np.int64(795)
```

In [19]:
```
sports_centre_energy_241UDS0111[0] * 2
```

Out[19]:
```
np.int64(1440)
```

In [20]:
```
sports_centre_energy_241UDS0111[1:3] / 2
```

Out[20]:
```
array([390. , 377.5])
```

In [ ]:
```
## Step 7 — Statistical Operations
```

In [21]:
```
library_energy_241UDS0111, computer_lab_energy_241UDS0111, sports_centre_energy_241UDS0111
```

Out[21]:
```
(array([860, 915, 890, 940]),
 array([1240, 1315, 1275, 1360]),
 array([720, 780, 755, 810]))
```

In [22]:
```
np.sum(library_energy_241UDS0111)
```

Out[22]:
```
np.int64(3605)
```

In [23]:
```
np.max(library_energy_241UDS0111), np.min(library_energy_241UDS0111)
```

Out[23]:
```
(np.int64(940), np.int64(860))
```

In [24]:
```
print(
    np.mean(library_energy_241UDS0111),
    np.mean(computer_lab_energy_241UDS0111),
    np.mean(sports_centre_energy_241UDS0111)
)
```

```
901.25 1297.5 766.25
```

In [ ]:
```
## Step 8 — Combine All Facility Data
```

In [25]:
```
library_energy_241UDS0111.ndim, computer_lab_energy_241UDS0111.ndim, sports_centre_energy_241UDS0111.ndim
```

Out[25]:
```
(1, 1, 1)
```

In [26]:
```
all_facility_energy_241UDS0111 = np.concatenate([
    library_energy_241UDS0111,
    computer_lab_energy_241UDS0111,
    sports_centre_energy_241UDS0111
])
all_facility_energy_241UDS0111
```

Out[26]:
```
array([ 860,  915,  890,  940, 1240, 1315, 1275, 1360,  720,  780,  755,
        810])
```

In [27]:
```
np.sum(all_facility_energy_241UDS0111)
```

Out[27]:
```
np.int64(11860)
```

In [ ]:
```
## Step 9 — Reshape into a 3 × 4 Matrix
```

In [28]:
```
all_facility_energy_241UDS0111.ndim, all_facility_energy_241UDS0111.shape
```

Out[28]:
```
(1, (12,))
```

In [29]:
```
all_facility_energy_241UDS0111 = all_facility_energy_241UDS0111.reshape(3, 4)
all_facility_energy_241UDS0111
```

Out[29]:
```
array([[ 860,  915,  890,  940],
       [1240, 1315, 1275, 1360],
       [ 720,  780,  755,  810]])
```

In [30]:
```
all_facility_energy_241UDS0111.ndim, all_facility_energy_241UDS0111.shape
```

Out[30]:
```
(2, (3, 4))
```

In [ ]:
```
## Step 10 — Sort and Split the Array
```

In [31]:
```
all_facility_energy_241UDS0111
```

Out[31]:
```
array([[ 860,  915,  890,  940],
       [1240, 1315, 1275, 1360],
       [ 720,  780,  755,  810]])
```

In [32]:
```
all_facility_energy_241UDS0111[0]
```

Out[32]:
```
array([860, 915, 890, 940])
```

In [33]:
```
np.sort(all_facility_energy_241UDS0111, axis=1)
```

Out[33]:
```
array([[ 860,  890,  915,  940],
       [1240, 1275, 1315, 1360],
       [ 720,  755,  780,  810]])
```

In [34]:
```
np.sort(all_facility_energy_241UDS0111, axis=0)
```

Out[34]:
```
array([[ 720,  780,  755,  810],
       [ 860,  915,  890,  940],
       [1240, 1315, 1275, 1360]])
```

In [ ]:
```
## Step 11 — Split the Array
```

In [37]:
```
split_library_241UDS0111, split_computer_lab_241UDS0111, split_sports_centre_241UDS0111 = np.split(
    all_facility_energy_241UDS0111,
    [1, 2],
    axis=0
)
```

In [38]:
```
split_library_241UDS0111
```

Out[38]:
```
array([[860, 915, 890, 940]])
```

In [39]:
```
split_computer_lab_241UDS0111
```

Out[39]:
```
array([[1240, 1315, 1275, 1360]])
```

In [40]:
```
split_sports_centre_241UDS0111
```

Out[40]:
```
array([[720, 780, 755, 810]])
```

In [ ]:
```
## Step 12 — Retrieve Week 2 Energy Use of All Facilities
```

In [41]:
```
all_facility_energy_241UDS0111
```

Out[41]:
```
array([[ 860,  915,  890,  940],
       [1240, 1315, 1275, 1360],
       [ 720,  780,  755,  810]])
```

In [42]:
```
week_2_energy_241UDS0111 = all_facility_energy_241UDS0111[:3, 1:2]
print(week_2_energy_241UDS0111)
```

```
[[ 915]
 [1315]
 [ 780]]
```

In [ ]:
```
RESULT:-
NumPy was used to create and inspect arrays, access values through indexing and slicing, update values, perform
arithmetic and statistical operations, concatenate and reshape arrays, sort and split data, and retrieve Week 2
energy use. The separate CSV contains the new PIN-specific dataset.
```